Testklausur

① LU-Zerlegung für $A = \begin{pmatrix} 0 & 2 & 1 & -1 \\ -2 & 3 & 0 & 2 \\ 0 & 0 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{pmatrix}$

L, U, P

$$\begin{array}{cccc} 0 & 2 & 1 & -1 \\ -2 & 3 & 0 & 2 \\ 0 & 0 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array} \quad \underset{\Longleftrightarrow}{p_1 = 2} \quad \begin{array}{cccc} -2 & 3 & 0 & 2 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ 2 & 1 & 1 & 0 \end{array} \quad \Longleftrightarrow \quad \begin{array}{c|ccc} -2 & 3 & 0 & 2 \\ \hline 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ -1 & 4 & 1 & 2 \end{array}$$

$$\underset{\Longleftrightarrow}{p_2 = 2} \quad \begin{array}{cccc} -2 & 3 & 0 & 2 \leftarrow 4 \\ 0 & 2 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ -1 & 2 & -1 & 4 \end{array}$$

$$\underset{\Longleftrightarrow}{p_3 = 4} \quad \begin{array}{cccc} -2 & 3 & 0 & 2 \\ 0 & 2 & 1 & -1 \\ -1 & 2 & -1 & 4 \\ 0 & 0 & 0 & 1 \end{array}$$

$$L = \begin{pmatrix} 1 & & & 0 \\ 0 & 1 & & \\ -1 & 2 & 1 & \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad U = \begin{pmatrix} -2 & 3 & 0 & 2 \\ & 2 & 1 & -1 \\ & 0 & -1 & 4 \\ & & & 1 \end{pmatrix}$$

$$p = (2, 2, 4)$$

(2) $f(x,y) = \frac{x}{y}$

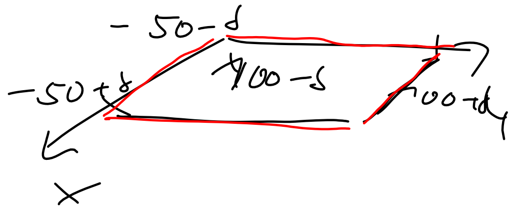
$x = -50 \pm \delta$

$y = 100 \pm \delta$ $\qquad \delta \leq 10$

a) absoluter Fehler:

$f(-50, 100)$ exakter Wert

$$e_a = \max_{\substack{x \in [-50-\delta, -50+\delta] \\ y \in [100-\delta, 100+\delta]}} | \overbrace{f(x,y) - f(-50,100)}^{g(x,y)} |$$



$\max(|g_{min}|, |g_{max}|)$

$= \max |g|$

$$g(x,y) = f(x,y) - f(-50,100)$$

$$= \frac{x}{y} - \frac{-50}{100}$$

$$= \frac{x}{y} + \frac{1}{2} \qquad \begin{cases} x \in [-60, -40] \\ y \in [90, 110] \end{cases}$$

$$\partial_x g(x,y) = \frac{1}{y} > 0$$

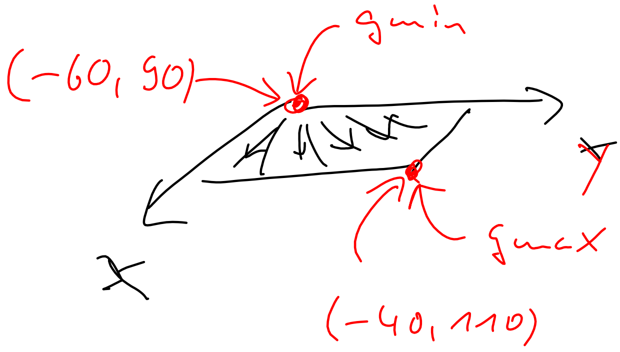
$$\partial_y g(x,y) = -\frac{x}{y^2} > 0$$

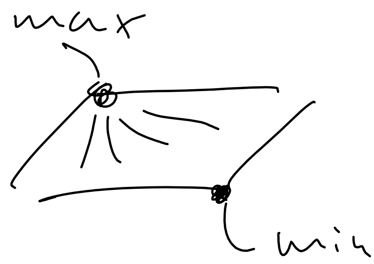
$\Rightarrow$ g monoton w. in x, y

auf $[-60, -40] \times [90, 110]$

$$g_{min} = g(-60, 90) = -\frac{1}{6}$$

$$g_{max} = g(-40, 110) = \frac{3}{22}$$





$$e_a = \max\left(|g_{min}|, |g_{max}|\right) = \max\left(\frac{1}{6}, \frac{3}{22}\right) = \frac{1}{6}$$

$$\boxed{\begin{aligned} f(x,y) &= \frac{x}{y} \\ \partial_x f &= \frac{1}{y} \\ \partial_y f &= -\frac{x}{y^2} \end{aligned}}$$

b) $$|f(x', y') - f(x,y)| \leq |\partial_x f(x,y)|\,|x'-x| + |\partial_y f(x,y)|\,|y'-y|$$

$$x = -50 \qquad y = 100$$

$$x' \in [-60, -40]$$
$$y' \in [90, 110]$$

$$\leq \left|\frac{1}{100}\right| 10 + \left|-\frac{-50}{100^2}\right| 10$$

$$= \frac{1}{10} + \frac{500}{10000} = \frac{1}{10} + \frac{5}{100} = \frac{15}{100} = 3/20$$

③ $A = \begin{pmatrix} 4 & 12 \\ -1 & 4 \end{pmatrix} \qquad b = \begin{pmatrix} 12 \\ 4 \end{pmatrix}$

a) $x^{(1)}, \dots, x^{(4)}$ für $x^{(0)} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Jacobi: $A = D - E - F$

$$Ax = b$$

$$\Leftrightarrow \left(D - (E+F)\right) x = b$$

$$\Leftrightarrow Dx = (E+F)x + b$$

$$x = D^{-1}(E+F)x + D^{-1}b$$

$$x^{(k+1)} = \underbrace{D^{-1}(E+F)}_{B_J} x^{(k)} + \underbrace{D^{-1}b}_{d_J}$$

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} \underline{a_{ij}}\, x_j^{(k)}\right)$$

$$A = \begin{pmatrix} \underline{4} & \underline{12} \\ \underline{-1} & \underline{4} \end{pmatrix} \qquad b = \begin{pmatrix} \underline{12} \\ \underline{4} \end{pmatrix}$$

$$x_1^{(k+1)} = \frac{1}{4}\left(12 - 12\, x_2^{(k)}\right) = \underline{3 - 3 x_k^{(2)}}$$

$$x_2^{(k+1)} = \frac{1}{4}\left(4 - (-1)\, x_1^{(k)}\right) = \underline{1 + \frac{1}{4} x_1^{(k)}}$$

k	0	1	2	3	4
$x^{(k)}$	$\begin{pmatrix} 0 \\ 0 \end{pmatrix}$	$\begin{pmatrix} 3 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 0 \\ \frac{7}{4} \end{pmatrix}$	$\begin{pmatrix} -\frac{9}{4} \\ 1 \end{pmatrix}$	$\begin{pmatrix} 0 \\ \frac{7}{16} \end{pmatrix}$

b) B_J, $\|B_J\|_\infty$, $\rho(B_J)$, Konvergenz?

$$B_J = D^{-1}(E+F)$$

$$A = D - E - F$$

$$A = \begin{pmatrix} 4 & 12 \\ -1 & 4 \end{pmatrix} = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} - \overbrace{\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}}^{E}$$

$$- \underbrace{\begin{pmatrix} 0 & -12 \\ 0 & 0 \end{pmatrix}}_{F}$$

$$B_J = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}^{-1} \underline{\underline{\begin{pmatrix} 0 & -12 \\ 1 & 0 \end{pmatrix}}}$$

$$= \begin{pmatrix} \frac{1}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix} \begin{pmatrix} 0 & -12 \\ 1 & 0 \end{pmatrix}$$

$$= \underline{\begin{pmatrix} 0 & -3 \\ \frac{1}{4} & 0 \end{pmatrix}} \leftarrow \begin{matrix} |0| + |-3| = 3 \\ |\frac{1}{4}| + |0| = \frac{1}{4} \end{matrix} \Big\} \max$$

$$\underline{\|B_J\|_\infty = 3}$$

$$\underline{\rho(B_J)} = \max_{k=1,2} |\underline{\lambda_k(B_J)}|$$

$$0 = \det(B_J - \lambda I) = \begin{pmatrix} -\lambda & -3 \\ \frac{1}{4} & -\lambda \end{pmatrix}$$

$$= \lambda^2 + \frac{3}{4}$$

$$\lambda^2 = -\frac{3}{4} \Rightarrow \lambda_{1,2} = \pm i \frac{\sqrt{3}}{2}$$

$$\left(|a+ib| = \sqrt{a^2+b^2}, \quad |ib| = |b| \right)$$

$$|\lambda_1| = |\lambda_2| = \frac{\sqrt{3}}{2} \Rightarrow \rho(B_J) = \frac{\sqrt{3}}{2}$$

Konvergenz: $\frac{\sqrt{3}}{2} < \frac{\sqrt{4}}{2} = 1 \Rightarrow$ Jacobi konv.

(4) $f(x) = \frac{1}{2}e^x - e^{-x}$, $X = [0,1]$

a) zeigen $f'(x) > 0$, $f''(x) = f(x)$:

$$f'(x) = \underline{\frac{1}{2}e^x + e^{-x}} > 0 \quad \text{da } e^y > 0 \ \forall y \in \mathbb{R}$$

$$\underline{f''(x)} = \frac{1}{2}e^x - e^{-x} = \underline{f(x)}$$

b) $g(x) = \frac{f(x)}{f'(x)}$ $\quad (\underline{\phi(x)} = x - g(x))$

g_{min}, g_{max} über $[0,1]$

$$g'(x) = \frac{f'(x)\,f'(x) - f(x)\,f''(x)}{(f'(x))^2}$$

$$\overset{a)}{=} \frac{(f'(x))^2 - (f(x))^2}{\underline{(f'(x))^2}}$$

Zähler: $(f'(x))^2 - (f(x))^2$

$$= \left(\frac{1}{2}e^x + e^{-x}\right)^2 - \left(\frac{1}{2}e^x - e^{-x}\right)^2$$

$$= \cancel{\frac{1}{4}e^{2x}} + 1 + \cancel{e^{-2x}} - \left(\cancel{\frac{1}{4}e^{2x}} - 1 + \cancel{e^{-2x}}\right)$$

$$= 2$$

$$g'(x) = \frac{2}{(f'(x))^2} > 0$$

$\Rightarrow$ g mon. w.

$$g_{min} = g(0) = -\frac{1}{3}$$

$$g_{max} = g(1) = \frac{e^2-2}{e^2+2} \quad (<1)$$

c) ϕ - Newton, zeige dass $\phi'(x) = (g(x))^2$

$$\phi(x) = x - \frac{f(x)}{f'(x)}$$

$$\phi'(x) = 1 - \frac{f'(x)\, f'(x) - f(x) \overbrace{f''(x)}^{a) \,= f(x)}}{(f'(x))^2}$$

$$= 1 - \frac{(f'(x))^2 - (f(x))^2}{(f'(x))^2}$$

$$= \frac{\cancel{f'(x)^2} - \cancel{f'(x)^2} + f(x)^2}{f'(x)^2}$$

$$= \left(\frac{f(x)}{f'(x)}\right)^2$$

$$= g(x)^2$$

d) ϕ Kontraktion auf $[0,1]$ mit $\alpha \leq \left(\frac{e^2-2}{e^2+2}\right)^2$

ϕ Kontr. auf $X \subset \mathbb{R}^n$ abg.:

(i) $\phi(X) \subset X$

(ii) $|\phi(x) - \phi(\tilde{x})| \leq \alpha\, |x - \tilde{x}| \quad \forall\, x, \tilde{x} \in X$

$\alpha \in [0,1)$

(i) $\phi'(x) \overset{c)}{=} g(x)^2 \geq 0$

$$\left.\begin{array}{l} \phi_{min} = \phi(0) = \frac{1}{3} \\ \phi_{max} = \phi(1) = \frac{4}{2+e^2} < 1 \end{array}\right\} \phi([0,1]) \subset [0,1]$$

(ii) $|\phi(x) - \phi(\tilde{x})| = |\phi'(\xi)|\,|x - \tilde{x}| \quad \xi \in co(x, \tilde{x})$

$|\underline{\phi'(\xi)}| \leq \alpha < 1 \quad \forall\, \xi \in [0,1]$

$\phi'(x) \overset{c)}{=} g(x)^2$, $\quad g_{min} = -\frac{1}{3}$

$= \max(g_{min}^2, g_{max}^2) \quad g_{max} = \frac{e^2-2}{\underline{e^2+2}} = \beta$

$= \beta^2 = \left(\frac{e^2-2}{e^2+2}\right)^2 =: \alpha$

$x = \phi(x)$

$\phi'(x) = 0$	$\rightarrow$ Ord. 2
$\phi''(x) = 0$	$\rightarrow$ " 3
$\phi'''(x) =$	$\rightarrow$ 4

⑤

x_i:	-2	0	1	2
y_i:	4	0	1	-4

a) <u>Newton - IP</u>:

x_i	y_i			
$\underline{-2}$	4			
$\underline{0}$	0	$\frac{0-4}{0-(-2)} = -2$		
$\underline{1}$	1	$\frac{1-0}{1-0} = 1$	$\frac{1-(-2)}{1-(-2)} = 1$	
$\underline{2}$	-4	$\frac{-4-1}{2-1} = -5$	$\frac{-5-1}{2-0} = -3$	

$$\frac{-3 - 1}{2-(-2)} = -1$$

$$
\begin{aligned}
p(x) = \underline{4} &+ (-2)(x - \underline{x_0}) \\
&+ 1\,(x - \underline{x_0})(x - x_1) \\
&+ (-1)\,(x - x_0)(x - x_1)(x - \underline{x_2})
\end{aligned}
$$

$$
\begin{aligned}
= 4 &- 2(x+2) \\
&+ (x+2)\,x \\
&- (\underline{x}+2)\,\underline{x}\,(\underline{x}-1)
\end{aligned}
$$

b)

x_i	-2	0	1	2
y_i	4	0	1	-4

Appr. $p(x) = ax + bx^3$

$$\sum (p(x_i) - y_i)^2 \longrightarrow \min$$

$$\| \begin{pmatrix} a x_0 + b x_0^3 - y_0 \\ \vdots \\ a x_3 + b x_3^3 - y_3 \end{pmatrix} \|_2^2$$

$$= \| A \binom{a}{b} - y \|_2^2 \quad , \quad A = \begin{pmatrix} x_0 & x_0^3 \\ & \vdots \\ x_3 & x_3^3 \end{pmatrix}$$

$$A^T A \binom{a}{b} = A^T b$$

$$A = \begin{pmatrix} -2 & -8 \\ 0 & 0 \\ 1 & 1 \\ 2 & 8 \end{pmatrix} \quad , \quad y = \begin{pmatrix} 4 \\ 0 \\ 1 \\ -4 \end{pmatrix}$$

$$A^T A = \begin{pmatrix} -2 & 0 & 1 & 2 \\ -8 & 0 & 1 & 8 \end{pmatrix} \begin{pmatrix} -2 & -8 \\ 0 & 0 \\ 1 & 1 \\ 2 & 8 \end{pmatrix}$$

$$= \begin{pmatrix} 9 & 33 \\ 33 & 129 \end{pmatrix}$$

$$A^T b = \begin{pmatrix} -2 & 0 & 1 & 2 \\ -8 & 0 & 1 & 8 \end{pmatrix} \begin{pmatrix} 4 \\ 0 \\ 1 \\ -4 \end{pmatrix} = \begin{pmatrix} -15 \\ -63 \end{pmatrix}$$

$$\begin{pmatrix} 9 & 33 \\ 33 & 129 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} -15 \\ -63 \end{pmatrix}$$

Gauß
$\Longrightarrow$ $a = 2$, $b = -1$

$$p(x) = -x^3 + 2x$$

(6) $\underbrace{\int_{-2}^{2} \frac{9}{5x^2+8}}_{f(x)} dx$

a) einf. Simpson

$$S = \frac{b-a}{6}\left(f(a) + 4 f\left(\frac{a+b}{2}\right) + f(b)\right)$$

$$= \frac{4}{6}\left(f(-2) + 4 f(0) + f(2)\right)$$

$$f(0) = \frac{9}{8}, \quad f(-2) = f(2) = \frac{9}{5 \cdot 4 + 8} = \frac{9}{28}$$

$$S = \frac{2}{3}\left(\frac{9}{28} + 4 \cdot \frac{9}{8} + \frac{9}{28}\right)$$

$$= \frac{2}{3}\left(\frac{9}{14} + \frac{9}{2}\right)$$

$$= \frac{2}{3} \frac{9+63}{14}$$

$$= \frac{\cancel{2}}{\cancel{3}} \frac{\cancel{72}\ 24}{\cancel{14}\ 7} = \frac{24}{7}$$

b) G_2 : $\underline{[-1, 1]}$ $\quad x_i : -\sqrt{\frac{3}{5}}, 0, \sqrt{\frac{3}{5}}$

$\downarrow$ $\quad \beta_i : \frac{5}{9}, \frac{8}{9}, \frac{5}{9}$

$[-2, 2]$ $\quad \tilde{x}_i : -2\sqrt{\frac{3}{5}}, 0, 2\sqrt{\frac{3}{5}}$

$\quad \tilde{\beta}_i : \frac{10}{9}, \frac{16}{9}, \frac{10}{9}$

$$G_2(f) = \underline{\tilde{\beta}_0 f(\tilde{x}_0) + \tilde{\beta}_1 f(\tilde{x}_1) + \tilde{\beta}_2 f(\tilde{x}_2)}$$

$$= \ldots = 3$$

$$[-1, 1] \rightarrow [a, b]$$

$$x_i \rightarrow \frac{b-a}{2} x_i + \frac{b+a}{2}$$

$$\beta_i \rightarrow (b-a)\beta_i$$

c) Romberg Ordnung 4

$$\begin{array}{ll} T_1 & \\ & \searrow \\ T_2 & \rightarrow \ \underline{T} \end{array}$$

$\uparrow$ Ordnung 2 $\quad \uparrow$ Ordnung 4

$$\int_{-2}^{2} \underbrace{\frac{9}{5x^2+8}}_{f(x)} \, dx$$

$$T_1 = \frac{b-a}{2}\left(f(a)+f(b)\right) = \frac{4}{2}\left(f(-2)+f(2)\right) = \frac{9}{7}$$

$$T_2 = \frac{b-a}{4}\left(f(a)+2f\left(\tfrac{a+b}{2}\right)+f(b)\right) = \frac{2}{2}\left(\underline{f(-2)+2f(0)+f(2)}\right) = \frac{81}{28}$$

l_i	T_i
$\underline{4}$	$\frac{9}{7}$
$\underline{2}$	$\frac{81}{28}$

$$\underline{\frac{81}{28} + \frac{\frac{81}{28}-\frac{9}{7}}{\left(\frac{4}{2}\right)^2-1}} = \ldots = \frac{24}{7}$$